

BROOKINGS, OR, USA

WMO: 725985

Lat: 42.074N

Lon: 124.290W

Elev: 140

StdP: 99.66

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24267

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.9	2.1	-8.1	1.9	10.9	-5.2	2.5	10.2	11.0	11.7	9.2	10.7	0.9	50	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	8.6	30.0	17.3	27.0	16.1	23.9	15.3	18.6	26.9	17.6	23.8	16.8	21.9	3.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	11.9	19.1	15.7	11.3	18.4	14.9	10.7	17.6	52.9	27.0	49.9	24.0	47.4	21.9	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years					
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.3	6.0	DB	-1.7	35.5	1.3	3.2	-2.6	37.8	-3.4	39.6	-4.1	41.4	-5.1	43.7
			WB	-2.6	20.6	1.4	1.4	-3.7	21.5	-4.5	22.3	-5.2	23.1	-6.3	24.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	8.9	8.9	8.9	10.4	12.4	14.7	15.4	15.5	15.7	13.4	10.6	8.6
	DBStd	4.04	2.87	3.02	2.61	2.99	2.95	3.27	2.98	2.45	3.55	2.97	2.73	2.78
	HDD10.0	278	55	52	53	26	5	0	0	0	0	3	24	59
	HDD18.3	2388	293	266	293	238	187	121	102	97	98	158	233	302
	CDD10.0	989	20	19	18	39	79	141	168	169	170	108	42	16
	CDD18.3	58	0	0	0	2	3	12	12	7	17	4	0	0
	CDH23.3	758	0	1	1	20	54	162	180	92	208	39	1	0
	CDH26.7	288	0	0	0	3	17	54	89	31	86	8	0	0
(14) Wind	WSAvg	2.1	2.1	2.4	2.4	2.3	2.3	2.2	1.6	1.5	1.8	1.9	2.2	2.3
(15) Precipitation	PrecAvg	1955	325	253	231	181	98	47	9	18	30	122	279	372
	PrecMax	3161	668	542	439	426	249	154	44	82	137	229	667	787
	PrecMin	1351	99	85	107	62	0	3	0	0	0	8	86	109
	PrecStd	465	149	132	97	97	59	38	11	21	32	67	140	166
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.5	21.1	19.9	26.3	29.6	31.7	35.0	31.0	34.1	27.4	21.3	18.1
		MCWB	11.5	12.7	11.7	13.6	15.6	16.7	18.5	18.7	17.5	14.5	12.7	9.9
	2%	DB	17.3	17.5	16.7	21.0	24.1	27.8	28.0	25.7	28.7	23.7	18.0	15.9
		MCWB	10.0	10.7	11.3	12.8	14.6	16.7	17.4	16.8	17.0	13.8	11.7	10.2
	5%	DB	14.2	14.8	14.8	17.4	20.7	24.2	23.0	22.3	25.1	21.1	16.1	13.9
		MCWB	10.4	10.0	10.7	11.4	14.0	15.7	16.4	16.4	15.9	13.1	12.0	10.7
	10%	DB	12.8	12.9	12.9	15.1	17.6	21.1	20.0	19.7	21.6	18.4	14.2	12.7
		MCWB	10.5	10.2	10.1	10.8	12.9	14.7	15.7	15.9	15.2	12.6	11.8	10.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	13.9	14.1	15.1	17.6	19.1	20.6	19.5	20.1	17.7	15.7	14.0
		MCDB	15.9	16.9	16.8	22.7	25.4	28.5	29.8	28.0	30.0	22.2	18.0	14.7
	2%	WB	12.5	12.6	12.7	13.5	16.1	17.6	18.1	17.9	18.2	16.2	14.2	12.7
		MCDB	13.8	14.6	14.9	18.8	22.0	25.4	26.3	23.3	25.9	20.0	15.7	13.6
	5%	WB	11.9	11.5	11.7	12.4	14.5	16.4	17.0	16.9	17.1	14.9	13.3	12.0
		MCDB	13.2	13.4	13.5	15.9	19.4	23.0	22.7	21.1	22.3	18.4	14.7	12.7
	10%	WB	11.0	10.7	10.8	11.6	13.3	15.2	15.8	16.0	16.0	14.0	12.4	11.1
		MCDB	12.5	12.3	12.3	14.2	17.1	20.3	20.0	19.5	20.0	16.4	13.7	12.1
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	6.4	6.8	7.5	7.8	8.6	8.0	7.4	8.6	7.7	6.4	6.4
		MCDBR	9.8	9.7	10.1	12.0	13.8	15.2	14.6	13.2	15.4	12.8	9.4	8.2
	5% WB	MCWBR	5.9	5.4	6.0	6.6	6.9	7.1	6.8	6.5	6.1	6.1	5.2	5.0
		MCWBR	7.3	7.0	7.7	9.9	11.7	13.1	12.5	11.0	11.6	9.3	6.7	5.2
(40) Clear-Sky Solar Irradiance	taub		0.314	0.314	0.320	0.343	0.348	0.339	0.340	0.339	0.331	0.323	0.310	0.314
		taud	2.576	2.584	2.589	2.482	2.511	2.544	2.548	2.566	2.559	2.599	2.612	2.578
	Ebn at Noon		842	901	933	928	926	931	927	920	908	877	842	809
		Edh at Noon	68	79	87	104	104	101	100	95	89	76	65	63
(44) All-Sky Solar Radiation	RadAvg		1.71	2.51	3.67	5.13	6.45	7.14	7.30	6.19	4.96	3.23	1.92	1.40
	RadStd		0.26	0.35	0.39	0.46	0.39	0.43	0.32	0.35	0.29	0.30	0.23	0.26

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+2

Nomenclature: See separate page